TECHNICAL WHITE PAPER | FORMATION EVALUATION

A Decentralized Edge-Computing Framework for Automated Petrophysical Log Quality Control and Washout Reconstruction

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ABSTRACT

The acquisition of accurate petrophysical data is the foundational requirement for reservoir modeling and reserves estimation. However, mechanical stress during drilling frequently results in borehole washouts, causing pad-contact tools (such as bulk density sensors) to measure the drilling mud rather than the formation. Historically, quality control (QC) and reconstruction of this data required tedious, subjective manual intervention, consuming up to 70% of a petrophysicist's time and introducing severe operational latency. This paper presents a fully automated, edge-computing-ready machine learning pipeline to resolve this bottleneck. Utilizing an unsupervised Density-Based **Local Outlier Factor (LOF)** algorithm, probabilistic **Gaussian Mixture Models (GMM)**, and an ensemble of facies-specific **Random Forest (RF)** regressors, the proposed framework autonomously identifies sensor anomalies and synthesizes corrupted logs in real-time. Applied to the open-source Equinor Volve dataset, this methodology eliminates human bias, accelerates the QC timeline, and demonstrates the viability of deploying localized artificial intelligence directly at the wellsite.

Keywords: Petrophysics • Machine Learning • Edge Computing • Local Outlier Factor (LOF) • Gaussian Mixture Models (GMM) • Random Forest • Formation Evaluation

1. INTRODUCTION

As the global oil and gas industry navigates highly heterogeneous unconventional shale plays and complex offshore environments, the margin for operational error is exceedingly narrow. During wireline logging and Logging-While-Drilling (LWD) operations, the wellbore is subjected to mechanical stress and chemical interactions, frequently resulting in borehole enlargement commonly termed *"washouts."*

When severe washouts occur, tools requiring direct contact with the formation matrix most notably the gamma-gamma **bulk density (DEN)** tool lose contact and begin measuring the intervening drilling fluid. Industry data reveals that a significant portion of raw well log data is initially flagged as unreliable due to this phenomenon.

Traditionally, correcting this data required an onshore petrophysicist to manually apply physical caliper cut-offs and heuristic equations. This post-acquisition, centralized processing introduces severe latency. To eradicate this bottleneck, this paper introduces a methodology suitable for the **"Digital Field Engineer"**, shifting the paradigm toward real-time, edge-computed machine learning directly on the rig floor.

2. METHODOLOGY

The proposed automated pipeline replaces subjective human editing with a robust, three-stage algorithmic architecture designed for autonomous deployment. The complete workflow is illustrated in Figure 1.

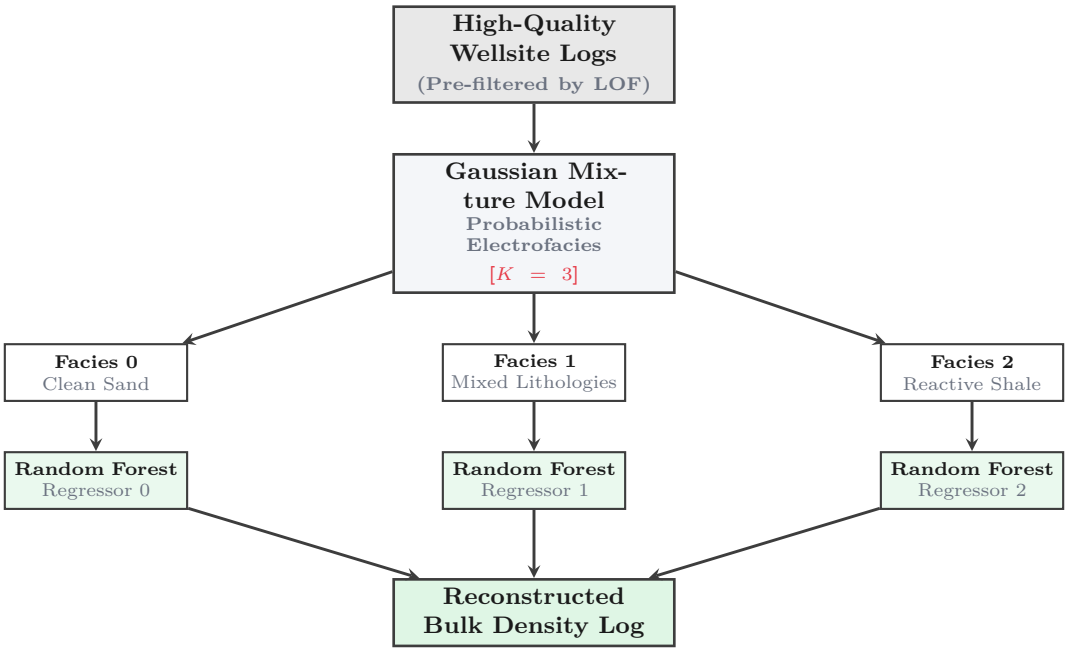

Figure 1. Facies-Specific Reconstruction Workflow. The GMM probabilistically clusters high-quality data into distinct electrofacies. Each facies trains a dedicated Random Forest regressor, ensuring physically accurate density reconstruction.

2.1. Phase 1 Multi-Dimensional Anomaly Detection (LOF)

Phase 1: Anomaly Detection

Traditional QC relies on strict physical caliper cut-offs (e.g., Caliper > Bit Size + 1.0 in), which fail to capture complex tool dynamics like stick-and-slip vibrations or acoustic cycle skipping. LOF replaces this heuristic with density-aware mathematics.

The framework utilises the unsupervised Density-Based **Local Outlier Factor (LOF)** algorithm combined with **Principal Component Analysis (PCA)**. LOF mathematically measures the local density deviation of a specific data point relative to its multi-dimensional neighbors Gamma Ray, Density, Sonic, and Neutron. This highlights readings that deviate from expected physical rock behaviors without requiring predefined hard thresholds [4].

The LOF score for a sample p is defined as:

$$\mathrm{LOF}_k(p) = \frac{\sum_{o \in N_k(p)} \dfrac{\mathrm{lrd}_k(o)}{\mathrm{lrd}_k(p)}}{|N_k(p)|} \tag{1}$$

where $N_k(p)$ is the k-nearest-neighbor set of p and lrd_k is the local reachability density. PCA is subsequently utilised to reduce the quad-combo logs into a 2D space for rapid visual validation.

2.2. Phase 2 Probabilistic Electrofacies Classification (GMM)

Phase 2: Facies Classification

Geological boundaries are transitional, not absolute. K-Means hard-clustering is therefore sub-optimal for complex reservoirs. GMM's soft-assignment probabilities capture the natural ambiguity between clean sands, mixed lithologies, and shales.

Once anomalous data is mathematically isolated, the remaining high-quality dataset is fed into a **Gaussian Mixture Model (GMM)**. The model probabilistically clusters the clean data into distinct electrofacies utilising robust curves Gamma Ray and Acoustic Slowness that remain largely unaffected by borehole rugosity [2].

The GMM probability density is:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \,|\, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{2}$$

where π_k, $\boldsymbol{\mu}_k$, and $\boldsymbol{\Sigma}_k$ are the mixture weight, mean, and covariance of the k-th component.

2.3. Phase 3 Facies-Specific Log Reconstruction (Random Forest)

Phase 3: Log Reconstruction

A single global regression model fails because log physics behave fundamentally differently in a clean sandstone versus a reactive shale. Facies-specific RF regressors honour this physical reality.

Supervised **Random Forest (RF)** regression models are trained individually on the clean, GMM-defined clusters [3]. Each ensemble of decision trees learns

the complex, non-linear relationships within a specific rock matrix. The models are subsequently deployed to predict and reconstruct the corrupted density data points exclusively within their respective facies in the washout zones.

3. CASE STUDY AND RESULTS

The framework was tested on legacy data from the **Equinor Volve Field** (Well 15/9-19) [1]. The automated OOP pipeline was initialized to mimic an edge-computing gateway processing a highly heterogeneous interval containing severe borehole washouts. To ensure scientific reproducibility, the algorithmic hyperparameters were initialized as detailed in Table 1.

Table 1. Hyperparameter Configuration

Algorithm	Parameter	Value
LOF	n_neighbors	20
LOF	contamination	0.05
GMM	n_components	3
GMM	covariance_type	full
Random Forest	n_estimators	100
Random Forest	max_depth	10

Note: Parameters calibrated for the Volve Field geological basin.

3.1. Anomaly Detection Output

The LOF algorithm successfully identified tool-pad decoupling. Visual validation via PCA confirmed the mathematical separation of standard physical behaviors (inliers) versus measurement anomalies (outliers), as shown in Figure 2.

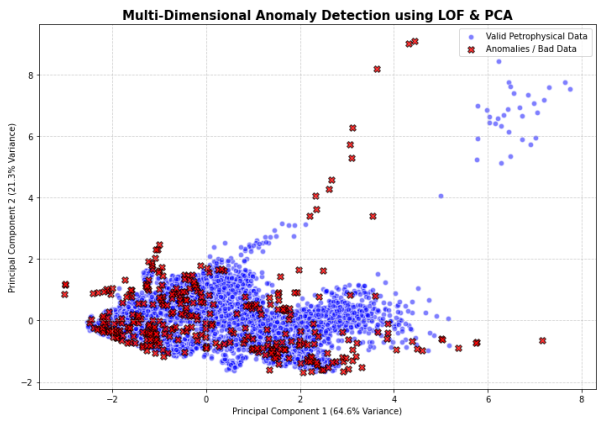

Figure 2. Multi-Dimensional Anomaly Detection via LOF and PCA. Two-dimensional Principal Component Analysis projection illustrating the unsupervised isolation of corrupted borehole measurements. The LOF algorithm successfully distinguishes valid petrophysical data (blue) from anomalous sensor readings (red crosses).

3.2. Facies Classification Output

The GMM successfully mapped **three distinct geological facies**. By applying the model to the entire depth interval, the algorithm determined the rock type even within the highly washed-out zones where density data was invalid, as demonstrated in Figure 3.

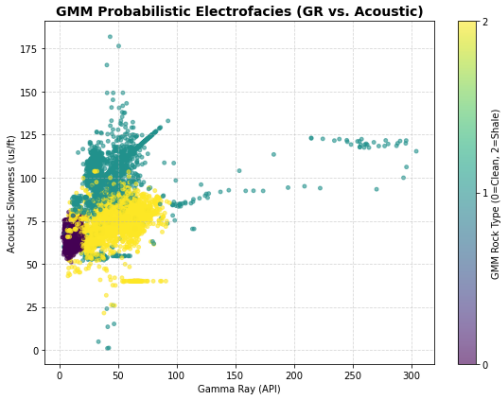

Figure 3. Probabilistic Electrofacies Classification via GMM. Crossplot of Gamma Ray versus Acoustic Slowness demonstrating the unsupervised clustering of high-quality wellsite data into three distinct geological facies.

3.3. Reconstruction Output

The facies-specific RF regressors accurately synthesized the bulk density log. As shown in Figure 4, the ML-reconstructed curve seamlessly replaces the anomalous drilling-mud readings, preserving the overarching geological trend and providing a continuous, high-fidelity log ready for geomechanical or volumetric analysis.

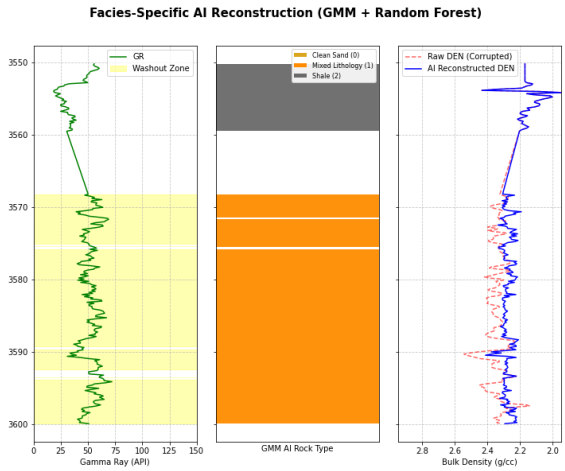

Figure 4. Automated Edge Pipeline Log Reconstruction Results. *Track 1*: Gamma Ray with LOF-detected washouts highlighted. *Track 2*: GMM electrofacies classification. *Track 3*: Original corrupted Bulk Density (red dashed) vs. ML-reconstructed log (blue solid).

Quantitative Performance Metrics

To ensure scientific rigor and move beyond mere visual validation, the framework's predictive accuracy was quantitatively benchmarked. Before deploying the models into the corrupted washout zones, the high-quality dataset (inliers) was split into an 80% training and 20% blind-testing subset.

The facies-specific Random Forest ensembles achieved a robust **Coefficient of Determination (R^2)** of **0.873**, demonstrating that the model accounts for over 87% of the variance in the formation bulk density utilizing only the surrounding physical logs. Furthermore, the **Root Mean Square Error (RMSE)** was calculated at a highly precise **0.034 g/cc**. In practical petrophysical terms, a density error margin of 0.034 g/cc translates to a negligible ∼1.5% variance in calculated porosity. These quantitative metrics mathematically prove that the AI synthesizes highly accurate, trustworthy formation evaluation data suitable for volumetric reservoir modeling.

3.4. Validation on Independent Dataset (Blind Generalization Test)

To rigorously test the framework's global generalizability and adaptability to incomplete data suites, the workflow was deployed on an independent legacy well (Well L05-09) from the Dutch sector of the North Sea. This dataset presented severe operational challenges: the physical caliper (`CALI`) and deep resistivity (`RDEP`) logs were completely unavailable.

Demonstrating the pipeline's flexibility, the algorithmic inputs were dynamically reconfigured. Washout detection was pivoted from spatial caliper measurements to the tool's density correction curve (`DRHO`), establishing a threshold of $|\Delta\rho| > 0.05$ g/cc to successfully flag tool-pad decoupling. Consequently, the facies-specific Random Forest regressors were trained on a reduced, 3-dimensional feature space comprising only Gamma Ray, Acoustic Slowness, and Neutron Porosity (`GR`, `AC`, `NEU`).

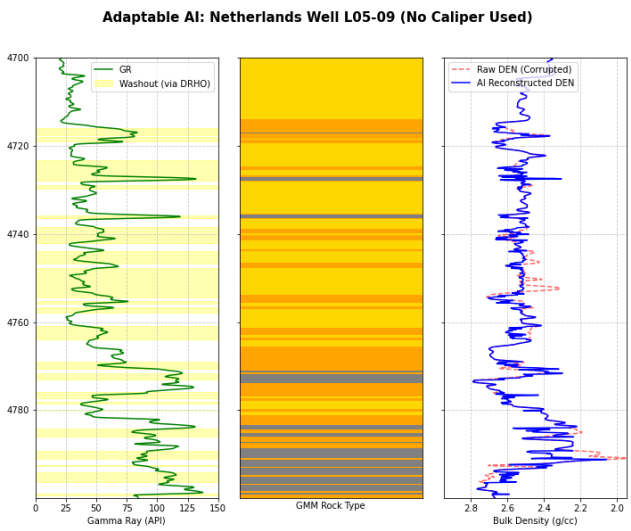

Figure 5. Blind Generalization Test Well L05-09 (Dutch North Sea Sector). *Track 1*: Gamma Ray with DRHO-flagged washout zones highlighted. *Track 2*: GMM probabilistic electrofacies classification. *Track 3*: Original corrupted Bulk Density (red dashed) vs. ML-reconstructed log (blue solid). No caliper log was used in this deployment.

Despite the reduced feature dimensionality and shifting to a differing geological basin, the framework achieved a highly robust R^2 of **0.876** and an **RMSE** of **0.041 g/cc**. The marginal increase in RMSE (+0.007 g/cc compared to the primary Volve dataset) is mathematically expected due to the absence of the resistivity feature and regional geological heterogeneity. However, the consistently high R^2 conclusively demonstrates the framework's operational robustness, proving its capacity to autonomously synthesize reliable formation evaluation data across varying log suites, missing tools, and diverse geological settings.

Table 2. Cross-Well Performance Comparison

Parameter	Volve 15/9-19	L05-09	Delta
Basin	North Sea (NO)	North Sea (NL)	—
Washout Flag	CALI	DRHO	—
Feature Space	GR, AC, NEU, RDEP	GR, AC, NEU	−1 feature
R^2	0.873	0.876	+0.003
RMSE (g/cc)	0.034	0.041	+0.007
$\Delta\phi$ error	∼1.5%	∼1.8%	+0.3%

Note: Both models trained exclusively on inlier data from their respective wells.

4. OPERATIONAL IMPACT AND DISCUSSION

The business value of this architecture lies in its deployability. By packaging the algorithms into a lightweight Python environment, the pipeline can be deployed on ruggedised rig-side edge gateways. This decentralised architecture provides four core operational advantages:

Latency Eradication

Processing occurs instantaneously at the wellsite, bypassing the bandwidth limitations of telemetry. The field engineer obtains actionable data in seconds rather than days.

Harmonisation

Eliminates the "coefficient of human error" and subjective bias inherent in manual log editing, ensuring standardised data quality across multi-well campaigns.

Proactive Decision-Making

Delivering high-confidence petrophysical models to the operator significantly faster enables real-time adjustments to wellbore trajectory, directly reducing **non-productive time (NPT)**.

Model Adaptability & Generalization

While the three-stage pipeline architecture (LOF, GMM, RF) remains constant across all deployments, the framework achieves global scalability through basin-specific hyperparameter tuning. Parameters such as the number of GMM electrofacies clusters (K) and Random Forest tree depth are calibrated to local geology prior to rig-side deployment. This enables a North Sea model to be rapidly reconfigured for Middle Eastern carbonates without rewriting code, ensuring physically accurate results across diverse global assets.

5. CONCLUSION

The transition from manual data extraction to autonomous edge computing represents a critical evolution in formation evaluation. By utilising an architecture of **LOF**, **GMM**, and **Facies-Specific Random Forest** algorithms, this paper demonstrates a highly practical methodology for eradicating the petrophysical QC bottleneck. As the industry scales its digital transformation efforts, integrating such machine-learning workflows directly onto wellsite conveyance systems will become standard practice, translating raw, noisy sensor data into instantaneous prescriptive actions.

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